Likely evaluator questions and grounded answers

HeatFlood Sentinel MS Data Science thesis defence · Aseef Ahmed (DS-029/2025/26) · Supervisor: Dr. Maria Andaleeb Siddiqui · NED UET

All figures come from the final thesis, the heatflood-sentinel repository results (results/*.json, results/tables, results/bigdata/*.json, RUN_LOG.md, DEVIATIONS.md) or the two cited government reports. Thesis results use ERA5 for 20 cities, 2015–2022, test year 2022. Big Data results were run on a **single-node pseudo-distributed cluster**. Defence date and examiner names: **[PLACEHOLDER — to be filled]**.

DATA AND GROUND TRUTH

Q1. Why did you use ERA5 reanalysis instead of PMD station data? Are the results valid for Pakistan?

PMD station data was requested in the proposal but no PMD data was obtained. The PMD adapter (file drop, column mapping, unit and range validation) is implemented but was exercised only with synthetic placeholder files. Every reported number therefore comes from ERA5 via the Open-Meteo archive (models=era5) at the nearest 0.25° cell for 20 Pakistani cities, 2015–2022. ERA5 is a physically consistent, gap-free reanalysis widely used for the region, but it smooths local extremes, so the results show skill *relative to ERA5*, not relative to station observations. Validation and bias correction against PMD records is the first item of future work, and I list it as the first limitation.

Evidence: DEVIATIONS.md D4, D5; thesis Ch. 6 Limitations and Future Work; Ch. 1 objective O3/O5.

Q2. Your flood labels are not observed floods. How can you claim flash-flood detection?

I do not claim flood detection accuracy. ERA5 does not observe floods, so both labels are rainfall-extreme proxies, and the thesis wording is “flash-flood-risk proxy derived from ERA5 precipitation extremes”. Proxy A (6-h rain ≥ max(50 mm, local p99) and antecedent precipitation index > p75) fires only 17 / 1 / 2 times in train / validation / test, too few to train or evaluate, so it is reported descriptively. The detectors use proxy B (trailing 24-h rain ≥ local p99.5 of daily totals; 929 / 79 / 405 positive windows). Validation against NDMA/PDMA and EM-DAT event records is future work.

Evidence: DEVIATIONS.md D1; thesis Ch. 3 hazard-definition table; results/bigdata/spark_etl_parity_2015_2022.json (label prevalence).

Q3. Why a percentile heatwave definition fitted on 2015–2020 rather than the WMO 1991–2020 baseline or an absolute 40 °C rule?

A relative (percentile) definition adapts to each city's climate, from Skardu to Jacobabad, so one rule works across all 20 cities: daily Tmax above the local calendar-day 95th percentile (±7-day window) on at least 3 consecutive days. Thresholds and climatologies are fitted on the training period only to prevent leakage within the short 2015–2022 window; using a 1991–2020 baseline would require the longer record. The absolute 40 °C variant is also computed (4,596 / 620 / 870 positive days). The 3-day duration reflects that the 2015 Karachi event was characterised by its persistence (MoCC 2015).

Evidence: Thesis Ch. 3 (climatology, hazard definitions); DEVIATIONS.md D5; MoCC Technical Report on Karachi Heat Wave, June 2015.

Q4. What is the status of the 1990–2026 data extension, and would it change your results?

It is in progress and is not used for any thesis number. When the Big Data run was made, the download was partial: 1990–1999 plus 2015–2022, i.e. 355 partitions and 3,111,696 hourly rows, loaded to HDFS (fsck HEALTHY). A Spark ‘lake’ ETL over these years (129,614 daily rows, 456 s on YARN) already shows why care is needed: pooling the 1990s into the climatology lowers the p95 thresholds, so relative-heatwave label days rise from 593 / 261 / 551 to 1,068 / 329 / 686 (train / val / test), while the absolute 40 °C label is unchanged. Thresholds must therefore be refitted and models retrained on the full, gap-free record before any comparison.

Evidence: results/bigdata/hdfs_load.json, spark_etl_lake.json; DEVIATIONS.md D8.

RESULTS AND EVALUATION

Q5. Your deep-learning models often do not beat the baselines. What is the contribution?

The contribution is an end-to-end, reproducible pipeline evaluated honestly against strong baselines, which the literature review shows is rare. Deep learning does win where the signal is smooth and spatial: the hourly LSTM beats XGBoost and persistence for temperature at every lead (RMSE 0.634 vs 0.679 vs 1.558 °C at +1 h), and the ConvLSTM beats persistence and climatology at every lead on the 2° lattice (1.262 vs 1.600 °C at +1 h; 2.287 vs 6.171 at +6 h). For daily Tmax, SARIMA is as good as or better than the LSTM at every lead, and I report that as a negative result rather than hiding it. RQ2 is answered “partly”.

Evidence: results/RESULTS.md; results/tables/leaderboard_hourly.csv, grid_convlstm_test.csv; thesis Ch. 6 RQ2 answer.

Q6. Why does a persistence rule (CSI 0.717) beat XGBoost (CSI 0.356) for next-day heatwaves, even though XGBoost has ROC-AUC 0.943?

Heatwaves are persistent by definition (≥3 consecutive hot days), so “tomorrow is a heatwave day if today is” is right most of the time: POD 0.835, FAR 0.165. XGBoost ranks days well (ROC-AUC 0.943, PR-AUC 0.589) but its decision threshold (0.325) was tuned on the milder 2021 season and transferred poorly to the exceptionally hot 2022, giving POD 0.407. The Spark MLlib GBT classifier reproduces this (CSI 0.367, ROC-AUC 0.947). The lesson is that persistence is the operational bar, and better operating thresholds (seasonal, POD-constrained) are needed.

Evidence: results/tables/detection_test.csv; results/bigdata/mllib_baseline.json; thesis Ch. 6 RQ3 answer and future work.

Q7. Isn't the flash-flood +6 h CSI of 0.609 just leakage?

Partly, and I say so on the slide. The +6 h target is the proxy-B flag of the next 6-h window, a trailing 24-h sum that shares 18 of its 24 hours with the tp24 feature available at issue time. The detector is therefore largely recognising rain that has already fallen, so 0.609 (vs 0.585 for persistence; PR-AUC 0.814) is an upper bound — nowcast or detection with overlap. The +24 h detector, whose target does not overlap the features, scores CSI 0 at the tuned threshold (PR-AUC 0.216, ROC-AUC 0.939). That is the honest forecast number.

Evidence: DEVIATIONS.md D1; results/tables/detection_test.csv.

Q8. Why do the detectors issue no alerts at longer leads (3-day heatwave, 24-h flood)?

Thresholds were tuned on the 2021 validation season and applied unchanged to 2022. At longer leads the predicted probabilities in 2022 never crossed the 2021-tuned thresholds (XGBoost heatwave 3 d threshold 0.15; flood 24 h 0.2), so POD and CSI are 0 even though the ranking skill is non-trivial (ROC-AUC 0.814 and 0.939). This is a threshold-transfer failure on an out-of-distribution year, which the thesis reports and addresses in future work through recalibration on recent data and POD-constrained thresholds.

Evidence: results/tables/detection_test.csv; thesis Ch. 5–6 (threshold transfer); Ch. 6 future work.

Q9. How did you establish statistical significance?

All models are scored on identical test samples. Paired Diebold–Mariano tests with block-bootstrap confidence intervals compare model pairs per lead, with Holm correction for multiple comparisons. For daily Tmax, SARIMA is significantly better than the LSTM at leads 1–3, XGBoost and SARIMA are not significantly different at lead 1, and the LSTM is significantly better than persistence from lead 3 onward. For daily precipitation the LSTM is significantly better than SARIMA and XGBoost at 3–7 days, but NSE stays near zero.

Evidence: results/tables/significance_daily.csv, significance_hourly.csv; thesis Ch. 3 evaluation design.

Q10. Why only one test year, 2022? Isn't that too little?

It is a limitation. 2022 was held out entirely (train 2015–2020, validation 2021) to avoid any temporal leakage. It is also an exceptional year — extreme pre-monsoon heat and the catastrophic monsoon floods (PDNA: 33 million affected, >1,730 deaths, damages >US$14.9 bn) — which makes it a stringent out-of-distribution test but not a representative one. Rolling-origin evaluation across several seasons, using the full ERA5 record, is listed as future work.

Evidence: Thesis Ch. 3 split; Ch. 6 Limitations and Future Work; PDNA 2022.

Q11. Some analyses use a single seed. How reproducible are the neural results?

The daily LSTM was repeated over three seeds (42, 7, 2024): Tmax RMSE 1.854 ± 0.015 °C at +1 day and 3.303 ± 0.085 at +7 days, so seed variance is small relative to model differences. The ConvLSTM (about 6 epochs, best epoch 3), XGBoost and the Spark MLlib models (seed 42) were run with one seed; I state this as a limitation. All code, configs and results are in the public repository with CI, so every number can be regenerated.

Evidence: Thesis table 'Daily LSTM: test RMSE over three random seeds' (results/tables/lstm_daily_seeds.csv); DEVIATIONS.md D2; results/bigdata/mllib_baseline.json.

Q12. Why LSTM and ConvLSTM rather than Transformers or global models such as GraphCast or Pangu-Weather? And where is the NWP comparison RQ2 asks for?

The approved proposal restricts the deep-learning prototype to LSTM and ConvLSTM (objective O4), which are well evidenced for station series and gridded fields and trainable on CPU with the available data. Global data-driven models are gap G5 and are future work. The NWP comparison could not be carried out because no archived NWP forecasts for 2022 were retrieved; the live site compares only with the current Open-Meteo forecast. RQ2 is therefore answered "partly", and I say so explicitly.

Evidence: Thesis Ch. 1 objectives; Ch. 2 gap table (G5); DEVIATIONS.md D7; Ch. 6 RQ2 answer.

BIG DATA AND STREAMING

Q13. You call it a Big Data system, but everything runs on one machine. Is it really distributed?

It is a single-node pseudo-distributed cluster and I describe it that way throughout. Every service and protocol is the real one — Hadoop 3.3.6 HDFS (NameNode + DataNode, replication 1), YARN (one NodeManager), Spark 3.5.9 on YARN in client mode, Kafka 3.9.2 in KRaft mode with one broker — on a 2-vCPU, 7.7 GB aarch64 host. It demonstrates that the pipeline runs end-to-end on real Big Data infrastructure (HDFS blocks, YARN containers, Kafka log, Spark state store and checkpoints). It does not demonstrate multi-node parallelism, cross-machine fault tolerance or network cost; that is future work. Nine YARN applications and 16 passing Big Data tests are logged.

Evidence: DEVIATIONS.md D8; results/bigdata/RUN_LOG.md; results/bigdata/streaming_kafka_spark.json (environment).

Q14. How do you know the Spark pipeline produces the same data as your pandas pipeline?

A parity check compares the Spark ETL output on YARN for 2015–2022 with the pandas reference, table by table. All 8 tables pass: silver hourly (1,402,560 rows) and daily (58,420), three climatologies, flash-flood 6-h labels (233,760), daily and hourly features. Row keys are identical, label, integer and string columns match exactly, and floats match within atol 1e-3 + rtol 1e-4; the largest difference is 1.7e-4 (heat index, float32 rounding). The only systematic difference is a provenance hash column, reported separately. The Spark ETL took 241.8 s on YARN.

Evidence: results/bigdata/parity_2015_2022.json, spark_etl_parity_2015_2022.json; DEVIATIONS.md D8.

Q15. Why did Spark MLlib's GBT do worse than XGBoost for next-day Tmax?

On the same 7,280 test rows, MLlib GBT reached RMSE 1.926 °C against 1.835 for XGBoost and 1.902 for persistence, so it does not beat persistence. MLlib's GBT lacks XGBoost's second-order gradients and regularisation, and it was run once with fixed parameters (100 iterations, depth 6, seed 42) and no tuning. For heatwave classification it matched XGBoost (CSI 0.367 vs 0.356). The purpose of the MLlib run was to show the model stage runs on YARN against the HDFS gold table, not to improve skill. (The thesis XGBoost figure of 1.822 °C uses a slightly different row set; 1.835 is the like-for-like number.)

Evidence: results/bigdata/mllib_baseline.json; results/RESULTS.md.

Q16. Explain your fault-tolerance result. What do exactly-once and at-least-once mean here, and where do the 42 duplicates come from?

In the kill-and-restart test a producer sent 21,600 events at 250 events/s; the Spark driver was SIGKILLed after 45 s, restarted 43 s later from the HDFS checkpoint, and drained the backlog in 34.9 s. The HDFS Parquet sink holds exactly 21,600 rows with 21,600 distinct Kafka partition-offset pairs: 0 missing, 0 duplicates, so that sink is effectively exactly-once (idempotent per micro-batch). Streaming features match an offline reference with max |diff| 0.0 and the 212 alerts are identical. The Kafka hf-alerts topic is at-least-once: it holds 254 messages, 42 of them replayed from the batch in flight at the crash, so consumers must de-duplicate by event id. The local runtime's kill-and-restart also lost and duplicated no alerts.

Evidence: results/bigdata/streaming_kafka_spark.json (kill_restart); results/streaming_local.json; RUN_LOG.md.

Q17. Are 2,739 events/s and a 5.5 s p95 latency good enough for early warning?

Yes for this workload: 20 cities produce 20 hourly observations per hour, so the requirement is minutes, not milliseconds. On the backlog test Spark processed 58,560 events in 3 micro-batches at 2,738.8 events/s (1,344 events/s including YARN start-up). At a steady 100 events/s the event-to-HDFS latency was p50 2.8 s, p95 5.5 s, p99 7.3 s, and alerts reached Kafka with p95 3.2 s. The local runtime handled a six-month season (87,840 events) at 210.5 events/s, about 37,900 times faster than real time. These are single-node figures and say nothing about scaling across nodes.

Evidence: results/bigdata/streaming_kafka_spark.json; results/streaming_local.json; thesis macros st.realtime_factor.

Q18. How does the live system avoid false alarms when it uses forecast data rather than ERA5?

The first live snapshot showed spurious heatwave emergencies for Sukkur and Larkana in late October because the recent days and 72-h forecast come from an NWP model running +2.0 to +2.6 °C warmer than ERA5 there, while thresholds are ERA5-based (train–serve skew). I added a per-city, per-UTC-hour bias correction estimated over the NWP–ERA5 overlap; Sukkur’s probability fell from 0.93 to 0.08. Karachi and Hyderabad rose because NWP was cold-biased there; under the relative definition these reflect an anomalously warm spell, and the page shows the probability, the p95 threshold and the forecast Tmax so users can judge.

Evidence: DEVIATIONS.md D3; live site https://siraseef.github.io/heatflood-sentinel/.

Q19. How would PMD or NDMA actually use this, and what is needed before deployment?

The code path (ingest, label, score, alert, dashboard/API) is in place and open source, and the live snapshot regenerates every 6 hours. Before operational use: (1) validate and bias-correct against PMD stations; (2) replace rainfall proxies with curated NDMA/PDMA flood events; (3) recalibrate alert thresholds seasonally with a POD constraint; (4) run the multi-broker Kafka and multi-worker Spark profiles and test node failure; (5) benchmark against archived NWP. These are the future-work items in Chapter 6.

Evidence: Thesis Ch. 6 Future Work; DEVIATIONS.md D6–D8; GitHub repository https://github.com/SirAseef/heatflood-sentinel.